



Modelling Financial Markets

Chapter 6: Multivariate Volatility and Dependence



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Acronyms used in this material

- ▣ AIC: Akaike Information Criterion
- ▣ ARCH: AutoRegressive Conditional Heteroskedasticity
- ▣ BAC: Bank of America Corporation (NYSE ticker)
- ▣ BEKK: Baba–Engle–Kraft–Kroner (multivariate GARCH model)
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BIC: Bayesian Information Criterion
- ▣ BNP: BNP Paribas SA (Euronext Paris ticker; Banque Nationale de Paris)
- ▣ CCC: Constant Conditional Correlation (model)
- ▣ CDO: Collateralised Debt Obligation
- ▣ CML: Canonical Maximum Likelihood
- ▣ DBK: Deutsche Bank AG (Xetra ticker)
- ▣ DCC: Dynamic Conditional Correlation
- ▣ EU: European Union
- ▣ EWMA: Exponentially Weighted Moving Average
- ▣ FRM: Financial Risk Meter
- ▣ GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ IFM: Inference Functions for Margins
- ▣ IIT: Illinois Institute of Technology
- ▣ JPM: JPMorgan Chase & Co. (NYSE ticker)
- ▣ US: United States
- ▣ VEC: VECtorised multivariate GARCH model (vech of the covariance matrix)

Outline

Why Correlations Matter

Rolling and EWMA Covariances

Multivariate GARCH and DCC

Crisis Correlations and the Forbes–Rigobon Bias

Copulas and Tail Dependence

Applications: Europe, Romania and Banks

Summary

References

Why Dependence Matters

Three reasons

- ▣ **Diversification:** portfolio risk depends on how assets move together
 - ▶ **Markowitz (1952):** variance of a portfolio is a quadratic form in the covariance matrix
- ▣ **Hedging:** the hedge ratio is a conditional regression slope
 - ▶ If the correlation changes, yesterday's hedge is wrong today
- ▣ **Contagion:** shocks spread across markets and institutions
 - ▶ Correlations tend to rise exactly when diversification is needed most

What we do in this chapter

- ▣ Estimate time-varying covariances: rolling windows, EWMA, multivariate GARCH, DCC
- ▣ Separate true contagion from the volatility bias (Forbes–Rigobon)
- ▣ Go beyond correlation: copulas and tail dependence
- ▣ Apply everything to stocks and bonds, Europe and Romania, Bitcoin and banks

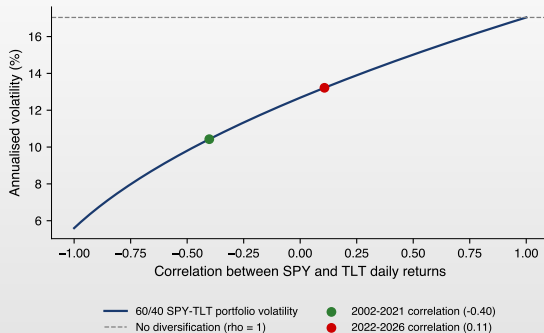
Diversification: The Algebra

- Portfolio weights $w \in \mathbb{R}^N$, covariance matrix Σ of the returns
- Portfolio variance: $\sigma_p^2 = w^\top \Sigma w = \sum_i \sum_j w_i w_j \rho_{ij} \sigma_i \sigma_j$
- Two assets:
 - ▶ $\sigma_p^2 = w^2 \sigma_1^2 + (1-w)^2 \sigma_2^2 + 2w(1-w)\rho \sigma_1 \sigma_2$
- Only $\rho = 1$ gives no risk reduction:
 $\sigma_p = w\sigma_1 + (1-w)\sigma_2$

SPY and TLT, 31 July 2002 – 2026

- SPY: S&P 500 exchange-traded fund; TLT: 20+ year US Treasury bond fund
- Annualised volatility: SPY 18.9%, TLT 14.3%
- 60/40 portfolio with $\rho = 1$: 17.0%; with $\rho = 0$: 12.7%
- With the correlation of 2002–2021 (-0.40): 10.4%
- With the correlation of 2022–2026 (0.11): 13.2%

The Value of a Negative Correlation



- Volatility of a 60/40 SPY-TLT portfolio as a function of the correlation, with the full-sample volatilities
- Points: the correlation measured before and after January 2022

Interpreting the Diversification Chart

- ▣ Moving from $\rho = -0.40$ to $\rho = 0.11$ raises the 60/40 volatility from 10.4% to 13.2%
- ▣ The volatilities of the two assets did not have to change: correlation alone does it
 - ▶ The bond sleeve stopped being a hedge and became a second source of risk
- ▣ The chart assumes one constant correlation per period; the rest of the chapter asks how to measure a moving one
- ▣ Lesson: a risk model with a fixed correlation can understate risk after a regime change

Hedging: The Minimum-Variance Hedge Ratio

- Hold one unit of asset S , sell h units of hedge instrument F : $r_H = r_S - h r_F$
- Minimise $\text{Var}(r_H) = \sigma_S^2 - 2h\sigma_{SF} + h^2\sigma_F^2$ over h
- Solution: $h^* = \sigma_{SF}/\sigma_F^2 = \rho\sigma_S/\sigma_F$, the slope of r_S on r_F
- Hedge effectiveness: share of variance removed $= \rho^2$

Worked example: BET hedged with Euro Stoxx 50

- Weekly returns, 10 January 2020 – 2026 ($n = 350$)
- $\hat{\sigma}_{BET} = 2.53\%$, $\hat{\sigma}_{SX5E} = 2.88\%$, $\hat{\rho} = 0.615$
- $h^* = 0.615 \times 2.53/2.88 = 0.540$
- Variance removed: $\hat{\rho}^2 = 0.378$; volatility falls by $1 - \sqrt{1 - \hat{\rho}^2} = 21.2\%$
- If ρ moves, h^* moves: a dynamic hedge needs a dynamic correlation

Contagion: When Correlations Jump



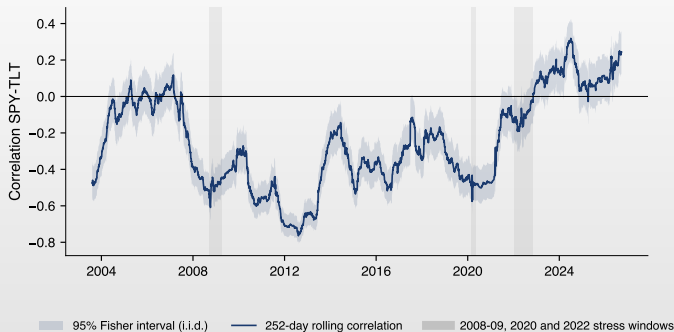
Lehman Brothers headquarters, New York, 15 September 2008, the day of the bankruptcy filing

Photo: Robert Scoble (2008); CC BY 2.0; Wikimedia

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- **Interdependence:** markets are linked in all states of the world
 - ▶ Trade, common investors, common macro factors
- **Contagion** ([Forbes & Rigobon \(2002\)](#)): a *significant increase* in cross-market linkages after a shock
 - ▶ A high correlation during a crisis is not enough; it must be higher than in calm times
- Section 4 shows why a naive comparison of correlations overstates contagion

Stocks and Bonds: A Correlation That Changed Sign



- ▣ Correlation of daily SPY and TLT returns on a 252-day window, with a 95% Fisher interval that assumes i.i.d. data
- ▣ Grey bands: autumn 2008 – March 2009, February–April 2020 and January–October 2022

Interpreting the Stock–Bond Correlation

- ▣ Minimum -0.76 (9 August 2012), maximum 0.32 (3 July 2024), last value 0.24
- ▣ Share of days with a positive rolling correlation: 6% before 2022, 81% since January 2022
- ▣ The sign depends on what drives bond yields
 - ▶ Growth shocks: stocks fall, yields fall, bonds rise (negative correlation)
 - ▶ Inflation shocks: stocks and bonds fall together (positive correlation), as in 2022
- ▣ The Fisher band is too narrow here: it ignores volatility clustering and the overlap of windows

Section Recap

- ▣ Portfolio risk is $w^\top \Sigma w$: the off-diagonal terms often dominate for large portfolios
- ▣ The minimum-variance hedge ratio is $\rho \sigma_S / \sigma_F$; its effectiveness is ρ^2
- ▣ Contagion means a significant *increase* of linkages, not just high co-movement
- ▣ The SPY–TLT correlation moved from mostly negative to mostly positive in 2022: correlations are not constants

Sample Covariance and Its Precision

- ▣ Returns $r_t \in \mathbb{R}^N$, $t = 1, \dots, T$; sample covariance $\hat{\Sigma} = \frac{1}{T-1} \sum_t (r_t - \bar{r})(r_t - \bar{r})^\top$
- ▣ Pearson correlation: $\hat{\rho}_{ij} = \hat{\Sigma}_{ij} / \sqrt{\hat{\Sigma}_{ii} \hat{\Sigma}_{jj}} \in [-1, 1]$
- ▣ Fisher transformation: $z = \text{artanh } \hat{\rho} \approx N(\text{artanh } \rho, 1/(T-3))$ for i.i.d. Normal data
 - ▶ 95% interval: $\tanh(z \pm 1.96/\sqrt{T-3})$
 - ▶ With $T = 252$ and $\rho = 0.5$ the half-width is about 0.09: a one-year correlation is noisy
- ▣ N assets: $N(N+1)/2$ distinct elements; with $N = 500$ this is 125 250 numbers

Rolling-Window Estimators

- ▣ Estimate on the last m observations: $\hat{\Sigma}_t = \frac{1}{m} \sum_{s=t-m}^{t-1} r_s r_s^\top$ (mean set to zero for daily data)
- ▣ Trade-off in m
 - ▶ Short window: reacts fast, but noisy
 - ▶ Long window: stable, but slow to detect a regime change
- ▣ **Ghost effect**: an extreme day enters with weight $1/m$ and leaves abruptly m days later
- ▣ Positive semidefinite by construction if $m \geq N$; singular if $m < N$

EWMA: The RiskMetrics Recursion

- Exponentially decaying weights:
 $\Sigma_t = \lambda \Sigma_{t-1} + (1 - \lambda) r_{t-1} r_{t-1}^\top$
- Equivalent: $\Sigma_t = (1 - \lambda) \sum_{k \geq 1} \lambda^{k-1} r_{t-k} r_{t-k}^\top$
- J.P. Morgan/Reuters (1996): $\lambda = 0.94$ for daily and $\lambda = 0.97$ for monthly data
- Half-life of a shock: $\ln 0.5 / \ln \lambda = 11.2$ days ($\lambda = 0.94$), 22.8 days ($\lambda = 0.97$)

Properties

- Positive semidefinite: a weighted sum of outer products
- No ghost effect: weights decay smoothly
- Same λ for all variances and covariances: very restrictive
- It is an integrated GARCH(1,1) without constant: no mean reversion (Chapter 5)

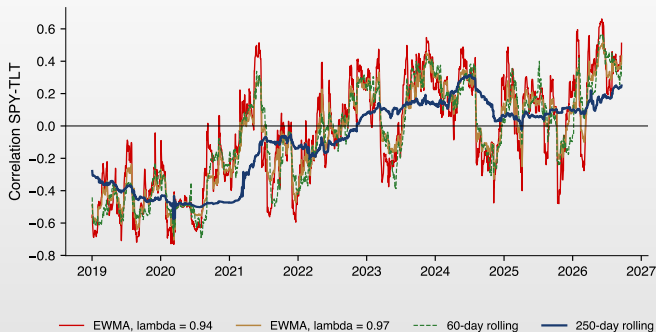
Worked Example: One EWMA Update, Step by Step

- SPY and TLT, returns in %; $\lambda = 0.94$
- Estimate for 17 September 2026: $\sigma_{11} = 0.395$, $\sigma_{22} = 0.307$, $\sigma_{12} = 0.139$
- Returns observed that day: $r_1 = 1.128$, $r_2 = 1.107$
- Correlation before the update:
 $0.139 / \sqrt{0.395 \times 0.307} = 0.400$

Update for 18 September 2026

- $\sigma_{11} = 0.94 \times 0.395 + 0.06 \times 1.128^2 = 0.448$
- $\sigma_{22} = 0.94 \times 0.307 + 0.06 \times 1.107^2 = 0.362$
- $\sigma_{12} = 0.94 \times 0.139 + 0.06 \times 1.128 \times 1.107 = 0.206$
- New correlation: 0.511
- A single day with SPY +1.13% and TLT +1.11% moved it from 0.40 to 0.51

EWMA versus Rolling Windows, SPY-TLT since 2019



- Two EWMA estimators ($\lambda = 0.94$ and 0.97) and two rolling windows (60 and 250 days) on the same data

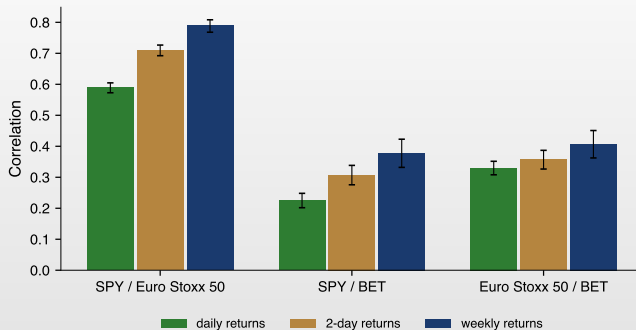
Interpreting the EWMA Comparison

- Standard deviation of the daily change in the estimate: 0.060 (EWMA 0.94), 0.032 (EWMA 0.97), 0.0089 (250-day window)
- EWMA with $\lambda = 0.94$ uses an effective sample of about $(1 + \lambda)/(1 - \lambda) \approx 32$ days: fast but noisy
- Persistently positive (20 days in a row) from 10 May 2022 with EWMA 0.94, but only from 8 December 2022 with the 250-day window
- Neither method estimates λ or m from the data: DCC (Section 3) does

Asynchronous Trading: A Trap for Daily Correlations

- Closing times differ: Bucharest and Frankfurt close hours before New York
 - ▶ US news after the European close enters European prices only the next day
- Consequence: same-day correlation is biased towards zero, and the lagged correlation is positive
- Remedies
 - ▶ Lower frequency: 2-day or weekly returns on common trading days
 - ▶ Add the lead-lag term: $\rho_0 + \rho_{+1}$ approximates the synchronous correlation
- Rule used throughout: join the *prices* on common days first, then compute returns

Daily, 2-Day and Weekly Correlations, 2000–2026



□ SPY, Euro Stoxx 50 and BET on common trading days; error bars: 95% Fisher intervals

Interpreting the Frequency Effect

- SPY–Euro Stoxx 50: 0.59 daily, 0.71 on 2-day returns, 0.79 weekly
- Lagged daily correlation (SPY yesterday, Euro Stoxx 50 today): 0.20; for BET: 0.17
- SPY–BET: 0.23 daily versus 0.38 weekly; Euro Stoxx 50–BET (almost synchronous closes): 0.33 versus 0.41
- The gap is largest across time zones: part of the “low” US–Europe daily correlation is a measurement artefact
- Weekly data are cleaner but give 1 367 instead of 6 428 observations: wider intervals

Section Recap

- ▣ Rolling windows: simple, but the window length is arbitrary and extreme days create ghost effects
- ▣ EWMA: $\Sigma_t = \lambda \Sigma_{t-1} + (1 - \lambda) r_{t-1} r_{t-1}^\top$, positive semidefinite, one decay for all elements, no mean reversion
- ▣ Sampling error of a correlation is large; Fisher intervals understate it under volatility clustering
- ▣ Across time zones use 2-day or weekly returns, or add lead-lag terms

From Univariate to Multivariate Volatility

- Chapter 5: $r_t = \sigma_t z_t$ with $\sigma_t^2 = \omega + \alpha r_{t-1}^2 + \beta \sigma_{t-1}^2$
- Now: $r_t = H_t^{1/2} z_t$, $\mathbb{E}(z_t) = 0$, $\text{Var}(z_t) = I_N$, so $\text{Var}(r_t | \mathcal{F}_{t-1}) = H_t$
- Any model for H_t must deliver
 - ▶ Positive definiteness at every t
 - ▶ Few parameters, so that estimation is feasible for many assets
 - ▶ Interpretable dynamics: volatility spillovers, time-varying correlation
- Gaussian quasi-log-likelihood: $\ell = -\frac{1}{2} \sum_t (N \ln 2\pi + \ln |H_t| + r_t^\top H_t^{-1} r_t)$

The VEC and BEKK Models

- **VEC** (Bollerslev, Engle & Wooldridge (1988)): every element of H_t depends on all squares and cross-products
 - ▶ $\text{vech}(H_t) = c + A \text{vech}(r_{t-1} r_{t-1}^\top) + B \text{vech}(H_{t-1})$
 - ▶ Positive definiteness is not guaranteed
- **BEKK** (Engle & Kroner (1995)): positive definite by construction
 - ▶ $H_t = CC^\top + A^\top r_{t-1} r_{t-1}^\top A + B^\top H_{t-1} B$
 - ▶ Parameters are hard to interpret; the likelihood is flat in high dimension

Number of parameters

N	VEC	full BEKK	DCC
2	21	11	$6 + 2$
5	465	65	$15 + 2$
10	6 105	255	$30 + 2$
50	3 252 525	6 275	$150 + 2$

- VEC: $N(N+1)/2 \cdot [1 + N(N+1)]$; BEKK: $N(N+1)/2 + 2N^2$
- DCC: $3N$ univariate GARCH parameters plus 2 (correlation targeted)

Constant and Dynamic Conditional Correlation

- Decomposition: $H_t = D_t R_t D_t$, $D_t = \text{diag}(\sigma_{1t}, \dots, \sigma_{Nt})$ from univariate GARCH
- CCC (Bollerslev (1990)): $R_t = R$ constant; standardised residuals $\varepsilon_{it} = r_{it}/\sigma_{it}$
- DCC (Engle (2002)): a GARCH-type recursion for a proxy matrix Q_t
 - ▶ $Q_t = (1 - a - b) \bar{Q} + a \varepsilon_{t-1} \varepsilon_{t-1}^\top + b Q_{t-1}$, $a, b \geq 0$, $a + b < 1$
 - ▶ $R_t = \text{diag}(Q_t)^{-1/2} Q_t \text{diag}(Q_t)^{-1/2}$
 - ▶ $\bar{Q}$: sample covariance of ε_t (correlation targeting)
- Two parameters govern all $N(N-1)/2$ correlations: parsimonious, but every pair has the same persistence

Robert Engle and the DCC Model



Robert F. Engle at the Western Economic Association
International conference, Santiago, 2017

Photo: [Econterms \(2017\)](#); CC BY-SA 4.0; [Wikimedia](#)

[Commons](#)

- Nobel Memorial Prize in Economic Sciences 2003 for ARCH models (Chapter 5)
- [Engle \(2002\)](#): “Dynamic Conditional Correlation”
 - ▶ Keeps the flexibility of univariate GARCH for each variance
 - ▶ Adds a simple, estimable law of motion for correlations
- Refinement: the corrected DCC of [Aielli \(2013\)](#), which makes $\bar{Q}$ a consistent target

Two-Step Estimation

- The Gaussian log-likelihood splits into a volatility part and a correlation part

$$\ell = \underbrace{-\frac{1}{2} \sum_t (N \ln 2\pi + 2 \ln |D_t| + r_t^\top D_t^{-2} r_t)}_{\ell_V} \underbrace{-\frac{1}{2} \sum_t (\ln |R_t| + \varepsilon_t^\top R_t^{-1} \varepsilon_t - \varepsilon_t^\top \varepsilon_t)}_{\ell_C}$$

- Step 1: fit N univariate GARCH(1,1) models; keep σ_{it} and ε_{it}
- Step 2: fix $\bar{Q}$ at the sample covariance of ε_t , maximise ℓ_C over (a, b)
- Inference
 - ▶ Consistent (quasi-maximum likelihood) even if z_t is not Normal
 - ▶ Standard errors from step 2 alone ignore the estimation error of step 1: too small
 - ▶ Test CCC against DCC: $H_0 : a = 0$; b is not identified under H_0 , so the χ^2 reference is only indicative

DCC Estimates for SPY and TLT

Step 1: GARCH(1,1), returns in %

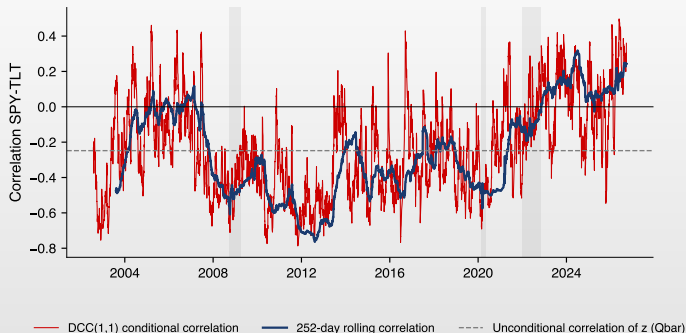
	ω	α	β	$\alpha + \beta$
SPY	0.0289	0.129	0.846	0.975
TLT	0.0057	0.049	0.944	0.993

Step 2: DCC(1,1)

	estimate	std. error
a	0.0529	0.0060
b	0.9326	0.0084
$a + b$	0.9856	
$\bar{Q}_{12}$	-0.248	

- Daily data 31 July 2002 – 18 September 2026, $T = 6\,023$
- Correlation persistence $a + b = 0.986$: half-life of a correlation shock ≈ 48 days
- Likelihood ratio DCC versus CCC: $2(\ell_C^{DCC} - \ell_C^{CCC}) = 526.4$
- Constant correlation is overwhelmingly rejected, whatever the exact reference distribution
- TLT volatility is more persistent (β larger) but reacts less to news (α smaller) than SPY

DCC Correlation of SPY and TLT, 2002–2026



- Conditional correlation $\rho_{12,t}$ from the DCC(1,1) model, the 252-day rolling correlation and the target $\bar{Q}_{12}$

Interpreting the DCC Path

- ▣ Range: from -0.79 (2 November 2011) to 0.50 (21 May 2026); last value 0.36
- ▣ Average DCC correlation: -0.54 from 15 September to 31 December 2008, -0.55 in 2012, -0.01 in 2022
- ▣ DCC reacts within days, like EWMA, but its speed (a) and memory (b) are estimated, not imposed
- ▣ Mean reversion to $\bar{Q}_{12} = -0.25$ is a weakness when the regime changes
 - ▶ Since January 2022 the path is above the target on 91% of days (36% before): a single long-run correlation is doubtful

Application: Minimum-Variance Weights with DCC

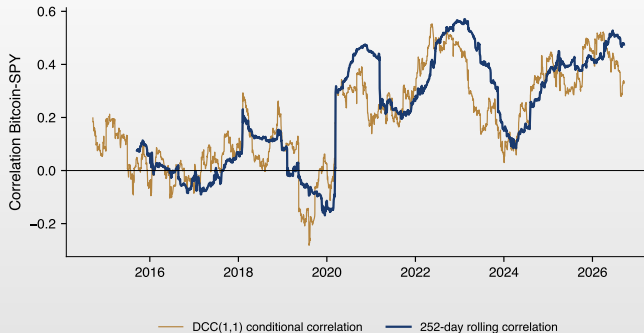


- Weight in SPY of the daily minimum-variance SPY–TLT portfolio: $w_t = (\sigma_{2t}^2 - \sigma_{12,t}) / (\sigma_{1t}^2 + \sigma_{2t}^2 - 2\sigma_{12,t})$, truncated to $[0, 1]$
- Same GARCH volatilities; correlation from DCC or constant (-0.30)

Interpreting the Portfolio Application

- Annualised volatility (%): 60/40 fixed 11.05; GARCH + constant correlation 9.15; GARCH + DCC 9.10
- Most of the gain comes from GARCH volatilities and minimum-variance weights; the DCC correlation adds only a small extra reduction
- Average absolute weight difference DCC minus constant: 0.019; largest: 0.22 (8 June 2026)
- Caveat: parameters are estimated on the full sample, so this is an in-sample illustration
 - ▶ A fair comparison re-estimates the models on expanding windows and adds transaction costs (Chapter 8)

Bitcoin and US Equities: DCC Correlation



- Bitcoin and SPY on common trading days since 18 September 2014 ($T = 3012$); weekend Bitcoin moves enter the Monday return

Interpreting the Bitcoin–Equity Correlation

- ▣ Average DCC correlation: 0.04 before 2020, 0.31 since January 2020; peak 0.55 on 19 May 2022
- ▣ Very persistent dynamics: $a = 0.015$, $b = 0.982$, $a + b = 0.9968$; CCC rejected (likelihood ratio 105.4)
- ▣ From diversifier to risk asset: since 2020 Bitcoin co-moves with equities much more than before
 - ▶ [Pele et al. \(2023\)](#) ask whether cryptos have become an asset class of their own; the rising correlation is part of the answer
- ▣ The full-sample correlation (0.24) hides two different regimes

Multivariate GARCH in Practice

- ▣ Large N : DCC with correlation targeting stays feasible; VEC and BEKK do not
 - ▶ For hundreds of assets, combine with factor models or shrinkage of $\bar{Q}$ (Chapter 3 and 4)
- ▣ Same a , b for all pairs: heterogeneous persistence needs richer (and harder) models
- ▣ Asymmetry: correlations rise more after joint falls; asymmetric DCC variants add a term for negative shocks
- ▣ Diagnostics: standardised residuals $\hat{H}_t^{-1/2} r_t$ should be serially uncorrelated with identity covariance
- ▣ Textbook treatment: [Franke, Härdle & Hafner \(2019\)](#), chapters on multivariate volatility and copulas

Section Recap

- ▣ $H_t = D_t R_t D_t$ separates volatilities (univariate GARCH) from correlations
- ▣ VEC and BEKK are general but explode in parameters; DCC needs $3N + 2$
- ▣ Two-step estimation is consistent; step-2 standard errors are too small
- ▣ SPY–TLT: $a + b = 0.986$, CCC rejected, sign change after 2022; Bitcoin–SPY: correlation up from 0.04 to 0.31

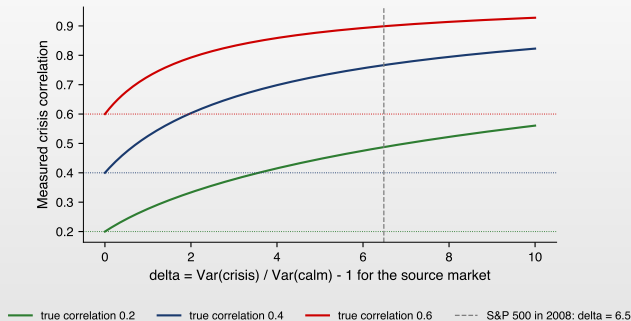
Do Correlations Rise in Crises?

- Common finding: correlations measured during crises are higher than in calm periods
- Two very different explanations
 - ▶ **Contagion**: the transmission mechanism itself changes
 - ▶ **Volatility bias**: the mechanism is unchanged, but a more volatile source market mechanically raises the correlation
- **Forbes & Rigobon (2002)**: after correcting the bias, most of the 1987, 1994 and 1997 “contagion” became interdependence
- **Longin & Solnik (2001), Ang & Chen (2002)**: correlations rise in bear markets, not in bull markets: an asymmetry the Normal distribution cannot produce

The Heteroskedasticity Bias: Derivation

- Model: $y_t = \alpha + \beta x_t + \varepsilon_t$, $\text{Cov}(x_t, \varepsilon_t) = 0$, β and σ_ε^2 constant
- Correlation: $\rho^2 = \frac{\beta^2 \sigma_x^2}{\beta^2 \sigma_x^2 + \sigma_\varepsilon^2} = \left(1 + \frac{\sigma_\varepsilon^2}{\beta^2 \sigma_x^2}\right)^{-1}$
- Crisis: $\sigma_{x,c}^2 = (1 + \delta) \sigma_x^2$ with $\delta > 0$; nothing else changes
- Then $\rho_c = \rho \sqrt{\frac{1 + \delta}{1 + \delta \rho^2}} > \rho$: the measured correlation rises although β is the same
- Inverting: $\rho^* = \frac{\rho_c}{\sqrt{1 + \delta(1 - \rho_c^2)}}$ is the crisis correlation adjusted for the volatility increase

How Much Can Volatility Alone Raise a Correlation?



- Measured crisis correlation ρ_c as a function of δ , for three true correlations that do not change
- Vertical line: the variance of 2-day S&P 500 returns was 7.5 times its calm-year level in the 2008 crisis window ($\delta = 6.48$)

The Forbes–Rigobon Test

- Procedure
 - ▶ Choose the source market x , a calm window and a crisis window
 - ▶ Compute $\delta = \hat{\sigma}_{x,c}^2 / \hat{\sigma}_{x,calm}^2 - 1$ and the adjusted ρ^*
 - ▶ Test $H_0 : \rho^* \leq \rho_{calm}$ with Fisher z statistics
- Use 2-day returns to reduce the time-zone problem (as in Forbes & Rigobon (2002))

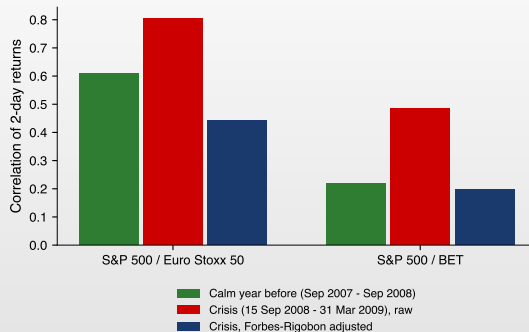
Assumptions to remember

- No omitted common shock: a global factor hitting both markets also raises σ_x
- No feedback from y to x (no endogeneity)
- Choice of windows matters: results can flip with a few weeks
- The correction can go too far if the true mechanism did change

Worked Example: The 2008 Crisis Step by Step

- Source market: S&P 500; calm year: 14 September 2007 – 12 September 2008 ($n = 241$ 2-day returns); crisis: 15 September 2008 – 31 March 2009 ($n = 130$)
- Variance ratio: $1 + \delta = 7.48$, so $\delta = 6.48$
- Euro Stoxx 50: $\rho_{calm} = 0.608$, $\rho_c = 0.804$
 - ▶ $\rho^* = 0.804 / \sqrt{1 + 6.48(1 - 0.804^2)} = 0.444$
 - ▶ One-sided p -value: 0.005 with ρ_c , 0.928 with ρ^*
- BET: $\rho_{calm} = 0.217$, $\rho_c = 0.485$, $\rho^* = 0.199$
 - ▶ One-sided p -value: 0.025 with ρ_c , 0.550 with ρ^*

2008: Raw versus Adjusted Crisis Correlations



- Correlation of 2-day returns with the S&P 500 in the calm year, in the crisis (raw) and after the Forbes–Rigobon adjustment

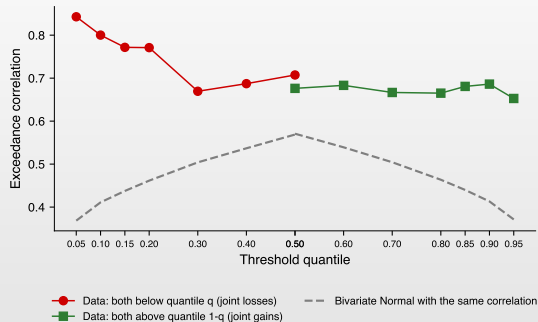
Interpreting the 2008 Test

- ▣ Raw correlations jump: Euro Stoxx 50 from 0.61 to 0.80, BET from 0.22 to 0.48; both increases significant at 5%
- ▣ After the adjustment neither increase survives: the 2008 co-movement is consistent with “no contagion, only interdependence”
- ▣ Why careful?
 - ▶ 2008 was a global shock that hit all markets at once: the “no common shock” assumption is doubtful
 - ▶ With $\delta \approx 6.5$ the correction is large, and the test has low power
- ▣ The seminar repeats the test for 2020 and checks the sensitivity to the windows

Exceedance Correlations

- ▣ Longin & Solnik (2001): correlation conditional on both returns being beyond a threshold
- ▣ Lower tail at quantile q : $\rho^-(q) = \text{Corr}(x, y \mid x \leq F_x^{-1}(q), y \leq F_y^{-1}(q))$
- ▣ Upper tail: the same with $x \geq F_x^{-1}(1 - q), y \geq F_y^{-1}(1 - q)$
- ▣ Benchmark: for a bivariate Normal distribution, $\rho^\pm(q) \rightarrow 0$ as $q \rightarrow 0$
 - ▶ Conditioning on extremes lowers the correlation even when nothing changes: compare with the Normal curve, not with ρ

Exceedance Correlations: S&P 500 and Euro Stoxx 50



- Weekly returns 2000–2026 ($n = 1\,393$, overall $\hat{\rho} = 0.78$); grey: bivariate Normal with the same correlation (2 million simulated pairs)

Interpreting the Exceedance Chart

- ▣ At $q = 0.10$: joint losses $\rho^- = 0.80$ (86 weeks), joint gains $\rho^+ = 0.69$ (70 weeks); Normal benchmark 0.41
- ▣ Both tails are far above the Normal benchmark: extremes are more dependent than a Gaussian model allows
- ▣ The lower tail is higher than the upper tail: crashes are shared more than rallies
- ▣ A correlation coefficient cannot describe this: we need a model for the whole dependence structure $\Rightarrow$ copulas

Section Recap

- ▣ Higher volatility of the source market raises the measured correlation even with an unchanged mechanism
- ▣ Adjusted correlation: $\rho^* = \rho_c / \sqrt{1 + \delta(1 - \rho_c^2)}$
- ▣ 2008: raw increases significant, adjusted increases not; the result rests on strong assumptions
- ▣ Exceedance correlations are asymmetric and far above the Normal benchmark

Why Correlation Is Not Enough

- ▣ Embrechts, McNeil & Straumann (2002): three fallacies about linear correlation
- ▣ Correlation measures only *linear* dependence
 - ▶ $x \sim N(0, 1)$ and $y = x^2$: $\rho = 0$, yet y is a function of x
- ▣ Correlation is not invariant to monotone transformations
 - ▶ $\text{Corr}(x, y) \neq \text{Corr}(e^x, e^y)$: prices and returns can show different correlations
- ▣ Margins and correlation do not determine the joint distribution
 - ▶ Two portfolios with identical margins and ρ can have very different Value-at-Risk (Chapter 7)
- ▣ Not every $\rho \in [-1, 1]$ is attainable for given margins: e.g. two lognormal variables

Sklar's Theorem



Main Building, Illinois Institute of Technology, Chicago: Abe Sklar (1925–2020) was professor of applied mathematics at IIT; no freely licensed portrait of him exists

Photo: Joe Ravi (2010); CC BY-SA 3.0; Wikimedia

[Commons](#)

- A **copula** is a joint distribution function on $[0, 1]^d$ with uniform margins
- Sklar (1959): for any joint distribution F with margins $F_1, \dots, F_d$ there is a copula C with
 - ▶ $F(x_1, \dots, x_d) = C(F_1(x_1), \dots, F_d(x_d))$
 - ▶ C is unique if the margins are continuous
- Conversely, any copula and any margins give a valid joint distribution
- Density: $f(x) = c(F_1(x_1), \dots, F_d(x_d)) \prod_i f_i(x_i)$

What the Copula Separates

- ▣ Margins: volatility, skewness, fat tails of each asset (Chapters 1 and 5)
- ▣ Copula: all the dependence, and nothing else; invariant to increasing transformations of each variable
- ▣ Three reference copulas in two dimensions
 - ▶ Independence: $\Pi(u, v) = uv$
 - ▶ Comonotonicity (perfect positive dependence): $M(u, v) = \min(u, v)$
 - ▶ Countermonotonicity: $W(u, v) = \max(u + v - 1, 0)$
- ▣ Fréchet–Hoeffding bounds: $W \leq C \leq M$ for every copula
- ▣ Reference text: [Nelsen \(2006\)](#)

Rank Correlations: Kendall and Spearman

- **Kendall's tau** (Kendall (1938)): concordant minus discordant pairs
 - ▶ $\tau = P[(X_1 - X_2)(Y_1 - Y_2) > 0] - P[(X_1 - X_2)(Y_1 - Y_2) < 0]$
 - ▶ Copula form: $\tau = 4 \iint C \, dC - 1$
- **Spearman's rho** (Spearman (1904)): Pearson correlation of the ranks
 - ▶ Copula form: $\rho_S = 12 \iint (C(u, v) - uv) \, du \, dv$
- Both depend only on the copula; both equal 1 for M , 0 for Π , -1 for W

Links to copula parameters

Copula	τ
Gaussian, t	$\frac{2}{\pi} \arcsin \rho$
Clayton	$\theta / (\theta + 2)$
Gumbel	$1 - 1/\theta$
Frank	$1 - \frac{4}{\theta} [1 - D_1(\theta)]$

- $D_1(\theta) = \frac{1}{\theta} \int_0^\theta \frac{t}{e^t - 1} dt$ (Debye function)
- Gaussian: $\rho_S = \frac{6}{\pi} \arcsin(\rho/2)$

Elliptical Copulas: Gaussian and Student t

- **Gaussian:** $C_{\rho}^{Ga}(u, v) = \Phi_{2,\rho}(\Phi^{-1}(u), \Phi^{-1}(v))$
 - ▶ Density: $c = \frac{1}{\sqrt{1-\rho^2}} \exp\left(-\frac{\rho^2(x^2+y^2)-2\rho xy}{2(1-\rho^2)}\right)$, $x = \Phi^{-1}(u)$, $y = \Phi^{-1}(v)$
 - ▶ Symmetric, no tail dependence for $|\rho| < 1$
- **Student t** (Demarta & McNeil (2005)): $C_{\rho,\nu}^t(u, v) = t_{2,\rho,\nu}(t_{\nu}^{-1}(u), t_{\nu}^{-1}(v))$
 - ▶ ν degrees of freedom: small ν = strong joint extremes; $\nu \rightarrow \infty$ gives the Gaussian copula
 - ▶ Symmetric tail dependence, equal in both tails
- Easy to extend to d dimensions with a correlation matrix: the standard choice in risk systems

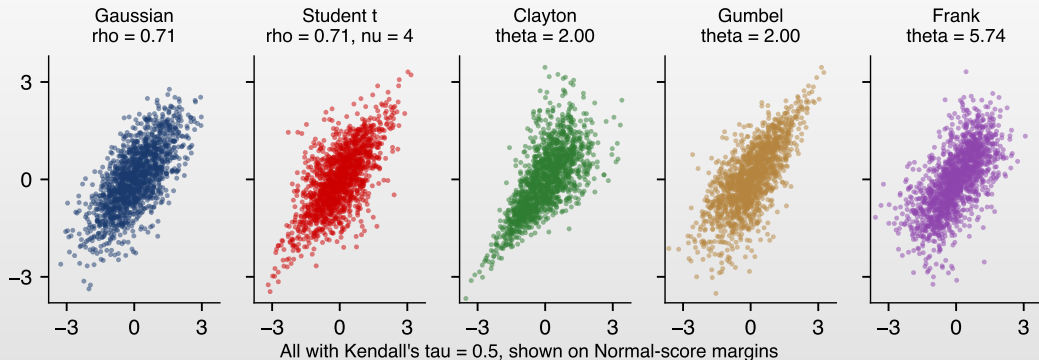
Archimedean Copulas: Clayton, Gumbel, Frank

- Built from a generator φ : $C(u, v) = \varphi^{-1}(\varphi(u) + \varphi(v))$
- **Clayton** (Clayton (1978)), $\theta > 0$
 - ▶ $C = (u^{-\theta} + v^{-\theta} - 1)^{-1/\theta}$
 - ▶ lower-tail dependence only
- **Gumbel** (Gumbel (1960)), $\theta \geq 1$
 - ▶ $C = \exp(-[(-\ln u)^\theta + (-\ln v)^\theta]^{1/\theta})$
 - ▶ upper-tail dependence only
- **Frank** (Frank (1979)), $\theta \neq 0$
 - ▶ $C = -\frac{1}{\theta} \ln\left(1 + \frac{(e^{-\theta u} - 1)(e^{-\theta v} - 1)}{e^{-\theta} - 1}\right)$
 - ▶ symmetric, no tail dependence

Why use them

- One parameter, closed-form C , easy simulation
- Asymmetric tails: Clayton for joint crashes of returns, Gumbel for joint extremes of losses
- Rotating a copula (using $1 - u, 1 - v$) moves its tail dependence to the other corner
- Limit: exchangeable, poor in high dimension (one parameter for all pairs)

Five Copulas with the Same Kendall's Tau



- 1500 simulated pairs from each copula with $\tau = 0.5$, plotted on Normal-score margins $(\Phi^{-1}(u), \Phi^{-1}(v))$
- Parameters from the τ table: Gaussian and t $\rho = 0.707$, Clayton and Gumbel $\theta = 2.0$, Frank $\theta = 5.74$

Interpreting the Copula Zoo

- ▣ Same Kendall's tau, same margins: the clouds differ only in the corners
- ▣ Gaussian and Frank: the corners thin out, so joint extremes are rare
- ▣ Student t: both corners stay dense
- ▣ Clayton: a sharp lower-left corner (joint crashes); Gumbel: a sharp upper-right corner
- ▣ A risk manager who calibrates only τ or ρ can pick any of the five and get very different tail risk

Tail Dependence

- Lower tail:
 $\lambda_L = \lim_{q \downarrow 0} P(V \leq q \mid U \leq q) = \lim_{q \downarrow 0} C(q, q)/q$
- Upper tail: $\lambda_U = \lim_{q \downarrow 0} P(V > 1 - q \mid U > 1 - q)$
- $\lambda > 0$: extremes arrive together no matter how extreme;
 $\lambda = 0$: asymptotic independence
- Tail dependence is a copula property, not visible in ρ

Closed forms

	λ_L	λ_U
Gaussian	0	0
Student t	$2 t_{\nu+1} \left(-\sqrt{\frac{(\nu+1)(1-\rho)}{1+\rho}} \right)$	same
Clayton	$2^{-1/\theta}$	0
Gumbel	0	$2 - 2^{1/\theta}$
Frank	0	0

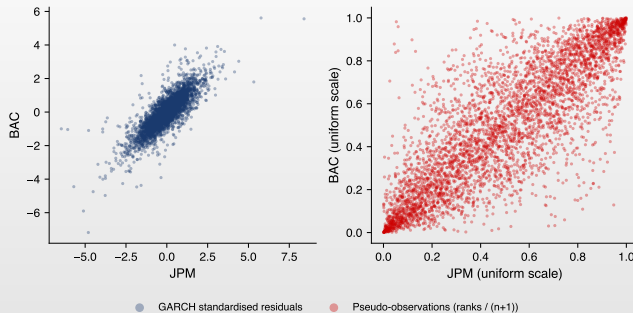
Worked Example: Tail Dependence Step by Step

- ▣ Student t with $\rho = 0.707$ and $\nu = 4$ (the zoo parameters)
 - ▶ Argument: $\sqrt{(\nu + 1)(1 - \rho)/(1 + \rho)} = \sqrt{5 \times 0.293/1.707} = 0.926$
 - ▶ Student t distribution function with $\nu + 1 = 5$ degrees of freedom: $t_5(-0.926) = 0.1984$
 - ▶ $\lambda_L = \lambda_U = 2 \times 0.1984 = 0.397$
- ▣ Clayton with $\theta = 2$: $\lambda_L = 2^{-1/2} = 0.707$; Gumbel with $\theta = 2$: $\lambda_U = 2 - \sqrt{2} = 0.586$
 - ▶ Same $\tau = 0.5$, yet the probability of a joint extreme ranges from 0 to about 0.7
- ▣ Meaning of $\lambda = 0.40$: given a far-tail crash in one asset, the other crashes too in about 40% of cases

Estimating Copulas

- ▣ **Full maximum likelihood**: margins and copula together; efficient but heavy and fragile
- ▣ **IFM** (inference functions for margins, [Joe \(2005\)](#)): two steps
 - ▶ Fit parametric margins (e.g. GARCH with Student t); set $\hat{u}_{it} = \hat{F}_i(x_{it})$; maximise $\sum_t \ln c(\hat{u}_{1t}, \hat{u}_{2t}; \theta)$
- ▣ **CML** (canonical maximum likelihood, [Genest, Ghoudi & Rivest \(1995\)](#)): no parametric margins
 - ▶ Pseudo-observations $\hat{u}_{it} = \text{rank}(x_{it})/(T + 1)$; then maximise the copula likelihood
 - ▶ Robust to misspecified margins; the $T + 1$ keeps $\hat{u}$ away from 0 and 1
- ▣ For returns: filter each series with GARCH first, then use the ranks of the standardised residuals
- ▣ Model choice: $\text{AIC} = 2k - 2\ell$, $\text{BIC} = k \ln T - 2\ell$

From Residuals to Pseudo-Observations: JPM and BAC



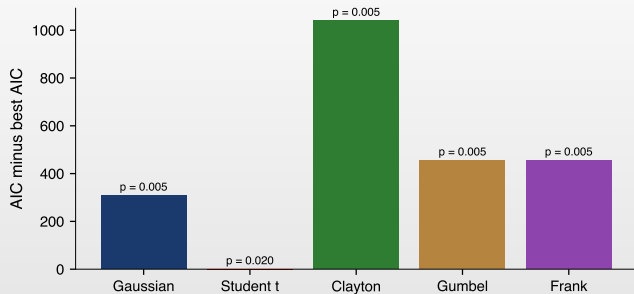
- JPMorgan Chase (JPM) and Bank of America (BAC), daily 2010–2026 ($T = 4\,122$), GARCH(1,1)-filtered
- Kendall's $\hat{\tau} = 0.644$, Spearman's $\hat{\rho}_S = 0.825$; both corners of the uniform scatter are dense

Copula Estimates for JPM and BAC

Copula	$\hat{\theta}$ or $\hat{\rho}$	$\hat{\nu}$	ℓ	ΔAIC	λ_L	λ_U	fit p
Gaussian	0.834	–	2449.3	308.3	0.000	0.000	0.005
Student t	0.844	4.28	2604.5	0.0	0.531	0.531	0.020
Clayton	2.394	–	2083.5	1039.9	0.749	0.000	0.005
Gumbel	2.596	–	2375.8	455.4	0.000	0.694	0.005
Frank	9.266	–	2375.8	455.4	0.000	0.000	0.005

- CML on pseudo-observations of GARCH residuals; ΔAIC relative to the best model
- Student t wins by far: $\hat{\nu} = 4.3$ means strong joint extremes, $\hat{\lambda} = 0.53$ in both tails
- Gaussian beats Clayton, Gumbel and Frank: the dependence is strong and roughly symmetric
- Last column: goodness-of-fit p -values (next slides)

Model Comparison and Goodness of Fit



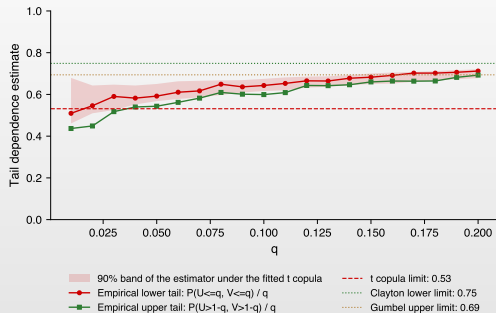
Numbers above bars: goodness-of-fit p-value (Rosenblatt Cramer-von Mises, parametric bootstrap)

- Bars: AIC distance to the best copula; numbers: bootstrap p -values of the goodness-of-fit test

Goodness of Fit: The Rosenblatt Test

- If $(U, V) \sim C_\theta$, then $E_1 = U$ and $E_2 = h(V | U) = \partial C_\theta(U, V) / \partial u$ are independent uniforms
- Statistic (Genest, Rémillard & Beaudoin (2009)): $S_n = n \int_{[0,1]^2} (D_n(e) - e_1 e_2)^2 de$, D_n the empirical distribution of (E_1, E_2)
- p -value by parametric bootstrap, because θ is estimated
 - ▶ Simulate n pairs from $C_{\hat{\theta}}$, convert to pseudo-observations, re-estimate, recompute $S_n^{(b)}$
 - ▶ $p = (1 + \#\{S_n^{(b)} \geq S_n\}) / (B + 1)$; here $B = 200$ (100 for Student t)
- Result: every family is rejected at 5%; Student t has by far the smallest statistic (0.099 versus 1.035 for the Gaussian)
- Why all rejected?
 - ▶ $T = 4122$ gives high power; a static copula over 16 years ignores that dependence moves (Section 3)
 - ▶ Remedy: time-varying (conditional) copulas, Patton (2006)

Empirical Tail Dependence: JPM and BAC



- Estimates $\hat{\lambda}_L(q) = \hat{C}(q, q)/q$ and $\hat{\lambda}_U(q)$ for q from 1% to 20%, with the 90% band of the same estimator on data simulated from the fitted t copula

Interpreting the Tail-Dependence Chart

- ▣ At $q = 5\%$: $\hat{\lambda}_L = 0.59$, $\hat{\lambda}_U = 0.54$; at $q = 1\%$: 0.51 and 0.44
- ▣ When JPM has one of its 5% worst days, BAC is in its own worst 5% about 59% of the time
- ▣ The lower-tail estimates lie inside the t-copula band (0.57–0.65 at 5%): the t copula describes the crashes well
- ▣ The upper tail is weaker: joint rallies are less tight than joint crashes, an asymmetry the t copula cannot produce
- ▣ Finite- q estimates are biased and noisy: λ is a limit, and very few points lie in the far corner

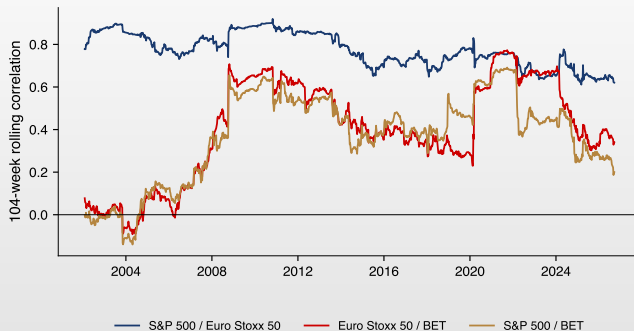
A Cautionary Tale: The Gaussian Copula and 2008

- ▣ Li (2000) modelled joint default times with a Gaussian copula
 - ▶ One correlation parameter, calibrated to market spreads, priced whole portfolios of credit
- ▣ It became the market standard for pricing tranches of collateralised debt obligations (CDO)
- ▣ What went wrong
 - ▶ Zero tail dependence: joint defaults in a crisis were treated as almost impossible
 - ▶ A static correlation calibrated in calm years: exactly the Forbes–Rigobon and DCC lessons
- ▣ The copula is a tool; the danger was a single, convenient, thin-tailed choice

Section Recap

- ▣ Sklar: joint distribution = margins + copula; the copula holds all the dependence
- ▣ Kendall's tau and Spearman's rho depend only on the copula and map one-to-one to one-parameter families
- ▣ Tail dependence: 0 for Gaussian and Frank, symmetric for t, lower for Clayton, upper for Gumbel
- ▣ Estimate on pseudo-observations of GARCH residuals; test fit with the Rosenblatt statistic and a parametric bootstrap
- ▣ JPM-BAC: t copula best, $\hat{\lambda} \approx 0.53$, but a static copula is still rejected

US, Euro-Area and Romanian Equities, 2000–2026

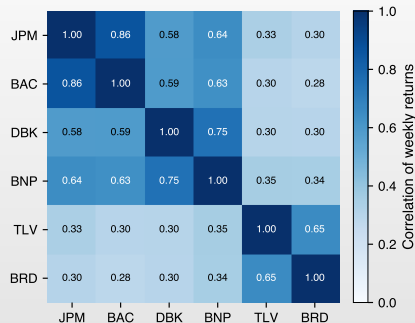


- Correlations of weekly returns on a 104-week window (common trading days, Friday closes)

Interpreting Market Integration

- ▣ Euro Stoxx 50–BET: 0.33 in 2000–2009, 0.45 in 2010–2019, 0.61 since 2020
- ▣ S&P 500–Euro Stoxx 50: 0.84, 0.78, 0.70 in the same periods
- ▣ Rolling Euro Stoxx 50–BET correlation: 0.31 at end-2007, 0.65 at end-2009, 0.29 at end-2019, 0.76 at end-2021, 0.35 now
 - ▶ Jumps in crises, partial return afterwards: integration or the Forbes–Rigobon volatility bias? The seminar tests it
- ▣ Lower correlation means better diversification for euro-area investors holding Romanian equities

Banks: Correlations of Weekly Returns, 2010–2026

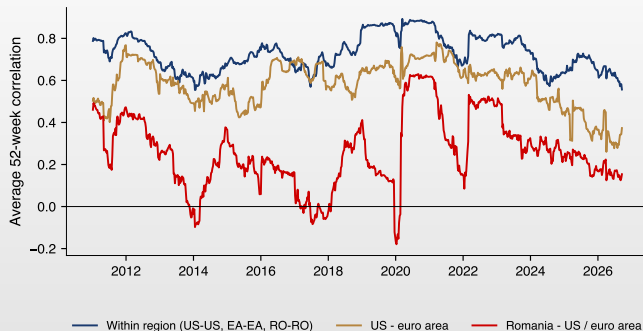


- US: JPM and BAC; euro area: DBK (Deutsche Bank) and BNP (BNP Paribas); Romania: TLV (Banca Transilvania) and BRD (BRD-Groupe Société Générale)
- 869 weeks from 15 January 2010

Interpreting the Bank Correlations

- ▣ Three blocks: US (0.86), euro area (0.75), Romania (0.65)
- ▣ Cross-region links are weaker for Romania: e.g. BNP–TLV 0.35, JPM–BRD 0.30
- ▣ BRD is controlled by Société Générale, yet its stock co-moves more with TLV than with any euro-area bank
- ▣ Bank returns share country-level factors: sovereign risk, domestic rates, regulation

Bank Correlations over Time

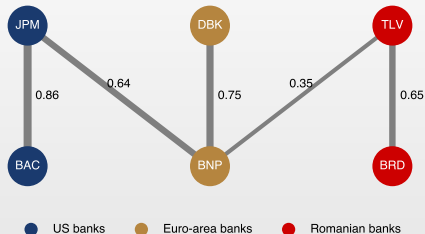


- Average of pairwise 52-week correlations of weekly returns within regions, between the US and the euro area, and between Romania and the others

Interpreting the Bank Correlations over Time

- ▣ Full-period averages: within regions 0.73, US–euro area 0.58, Romania–others 0.26
- ▣ March–June 2020: 0.87, 0.68 and 0.56: the Romanian link more than doubled in a global shock
- ▣ Latest values: 0.56, 0.37, 0.15
- ▣ Diversification across banking systems works in normal times and weakens exactly in systemic events

A Correlation Network of Banks



- Minimum spanning tree ([Mantegna \(1999\)](#)) on distances $d_{ij} = \sqrt{2(1 - \rho_{ij})}$: the 5 strongest links that connect all 6 banks without cycles
- Edges: JPM–BAC (0.86); JPM–BNP (0.64); DBK–BNP (0.75); BNP–TLV (0.35); TLV–BRD (0.65); Romania is attached through BNP–TLV

From Correlations to Systemic Risk

- ▣ The tree shows the path along which a shock is most likely to travel: US $\leftrightarrow$ euro area $\leftrightarrow$ Romania
- ▣ Correlation networks are the first step towards systemic-risk measures (Chapter 18)
 - ▶ Connectedness from Granger causality: [Billio, Getmansky, Lo & Pelizzon \(2012\)](#)
 - ▶ Spillover indices from variance decompositions: [Diebold & Yilmaz \(2009\)](#)
 - ▶ Tail-event networks: the Financial Risk Meter applied to EU energy firms, [Toma, Dima, Grecu, Pele & Lessmann \(2026\)](#)
- ▣ Energy prices and European stock markets: dependence across sectors, [Grecu, Cramer, Pele & Lessmann \(2024\)](#)
- ▣ Correlation networks inherit all the problems of correlations: non-linearity, time variation, tails

Section Recap

- ▣ BET–Euro Stoxx 50 weekly correlation rose from 0.33 (2000s) to 0.61 (since 2020)
- ▣ Bank returns cluster by region; Romanian banks are the least connected, except in global shocks
- ▣ Minimum spanning trees summarise a correlation matrix as a network, the bridge to Chapter 18

Common Pitfalls

- ▣ Computing returns before joining series with different calendars (spurious Monday returns, lost days)
- ▣ Comparing daily correlations across time zones without 2-day or weekly returns
- ▣ Calling a crisis increase “contagion” without the Forbes–Rigobon adjustment
- ▣ Reading copula fit only from AIC: the best of five can still be rejected
- ▣ Assuming a Gaussian copula for risk: zero tail dependence by construction
- ▣ Trusting two-step standard errors or Fisher intervals under volatility clustering

Where This Chapter Fits

Builds on

- ▣ Chapter 1: fat tails and volatility clustering of single series
- ▣ Chapter 4: portfolio optimisation needs Σ
- ▣ Chapter 5: univariate GARCH is step 1 of DCC and the filter before copulas

Leads to

- ▣ Chapter 7: portfolio Value-at-Risk and Expected Shortfall with DCC covariances and simulated copulas
- ▣ Chapter 8: backtesting multivariate risk forecasts
- ▣ Chapter 18: systemic risk and financial networks

Conclusions

- ▣ Correlations move: SPY–TLT changed sign, Bitcoin–SPY rose from about zero to 0.31, BET joined the European cycle
- ▣ EWMA is simple, DCC estimates the dynamics with $3N + 2$ parameters and rejects constant correlation
- ▣ Crisis correlations must be adjusted for volatility before speaking of contagion
- ▣ Copulas separate margins from dependence; tail dependence is what matters for joint losses
- ▣ Static models of dependence are rejected in long samples: dependence itself is dynamic

Further Study on Quantinar

Advanced courses on quantinar.com

- ▣  Measuring Statistical Risk
- ▣  Multivariate Statistical Analysis
- ▣  Minimum Spanning Tree
- ▣  FRM: Financial Risk Meter

Code, data and textbooks

- ▣ All charts and tables: github.com/danpele/MFM/tree/main/Quantlets/Ch_06
- ▣ Textbooks: Franke, Härdle & Hafner (2019); Härdle & Simar (2019); Nelsen (2006)

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